

Roof Open-Source Public Development Roadmap

Project Roof

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Roof is an open-source initiative by Data Foundation Lab to develop the foundations of a Personal Internet. An environment in which individuals can own and operate their personal computing and data infrastructure, access applications, and communicate and share data without depending entirely on centrally operated platforms.

Roof was initially developed around dedicated Raspberry Pi hardware, with Roof.Engine providing the underlying infrastructure and RoofDeck serving as the desktop client.

While the existing implementation already includes foundational capabilities for identity, communication, storage, networking and application integration, parts of the system are closely coupled to the assumptions of operating a dedicated Linux-based device.

Our next major development objective is to change that.

We are transitioning Roof towards a hardware-independent, software-first platform.

Our intention is for individuals to operate Roof on supported computing devices they already own, without needing to purchase, configure or maintain dedicated Raspberry Pi hardware.

Dedicated hardware will remain a possible deployment option, rather than a prerequisite for using Roof.

This roadmap outlines the development priorities, planned milestones and technical direction for that transition.

Current Development Principles

Our current development follows several guiding principles:

1. Hardware independence is primarily an architectural separation and adaptation effort, not a complete rewrite of Roof.
2. The core Roof runtime should operate without requiring a dedicated Raspberry Pi or ownership of an entire host operating system.
3. Users should be able to operate Roof on supported general-purpose computing devices, with dedicated hardware as an optional deployment method.
4. Running Roof on a different processor architecture or operating system does not automatically mean it is safe or ready for everyday use.
5. Roof should not perform dedicated-appliance administration or modify system-wide settings on general-purpose devices without explicit configuration and appropriate safeguards.
6. Existing functionality, security and reliability must be demonstrated on the independent runtime before introducing major new capabilities.
7. Progress, verified support, limitations and opportunities for contribution should be made visible to the community.

Initial Development Roadmap

Period	Focus	Target milestone
Nov 2026	Initial source publication	Developer Preview 0.1.0
Nov-Dec 2026	Dependency assessment and architectural separation	Hardware-Independence Architecture Baseline
Dec 2026	Generic Linux runtime proof of concept	Hardware-Independent Runtime PoC
Jan 2027	Safe local software runtime	Independent Runtime Alpha (Local)
Feb 2027	Independent-node networking and recovery	Independent Runtime Alpha (Remote)
Mar 2027	Existing application, storage and sharing workflows	Functional Integration Review
Apr 2027	Safe installation on general-purpose devices	Software-First Installation Alpha
May-Jun 2027	Core functionality on supported user devices	Roof Software Beta 1
Jul-Sep 2027	Additional platform evaluation and expanded validation	Public Software Beta Candidate
Oct-Dec 2027	Platform consolidation and optional hardware	2027 Platform Review

All dates are indicative development targets. Releases and changes to supported environments are subject to technical validation, security review and release-readiness criteria.

Q4 2026: Open-Source Publication and Hardware-Independence Foundations

Period: November-December 2026

Developer Preview 0.1.0 (November)

Target Release: 16 November 2026
Release Type: Experimental, source-only developer preview

Our first public milestone is the publication of Roof's existing source code.

Developer Preview 0.1.0 will enable external developers and researchers to inspect the implementation, understand its architecture, independently build supported projects and begin contributing to development.

Planned Deliverables

- Publish the approved Roof and RoofDeck source repositories.
- Publish the applicable open-source licences and third-party notices.
- Provide project architecture and component documentation.

- Publish development prerequisites and source-build instructions.
- Document supported development environments, infrastructure requirements and existing hardware dependencies.
- Provide development configuration guidance and appropriate infrastructure separation.
- Document known defects, unsupported functionality and implementation limitations.
- Establish public issue tracking, contribution guidelines and a private security vulnerability reporting process.

Important Release Limitations
Developer Preview 0.1.0 is a source-code release, not a consumer software beta.

It will not provide prebuilt executable installers, a supported general-purpose-device deployment workflow or a completed hardware-independent Engine.

The existing Engine contains Linux-specific and privileged host-management functionality originally designed for dedicated Roof hardware. Attempting to operate it on an ordinary computer without appropriate isolation may affect host-system configuration.

The initial public release is intended for source inspection, development and technical evaluation.

Success Criteria: External developers can access the approved source, build the documented supported implementation, review its limitations and participate through the published contribution channels.

Hardware-Independence Assessment (November-December)

Target Milestone: Hardware-Independence Architecture Baseline
Target Date: 4 December 2026

Following source publication, we will assess the existing Engine architecture to distinguish core Personal Internet functionality from hardware-specific and privileged operating-system management.

The primary objective is to preserve established Roof functionality while removing the assumption that every Roof deployment controls a dedicated Linux appliance.

Planned Work

- Identify Raspberry Pi-specific, processor architecture-specific and Linux-specific dependencies.
- Identify privileged host operations and dedicated-appliance administration requirements.
- Separate core Engine responsibilities from optional device provisioning and management.
- Define the minimum runtime required for hardware-independent Roof operation.

- Establish explicit deployment profiles for software-based hosts and dedicated appliances.
- Design data isolation without requiring a dedicated Linux operating-system account for every client.
- Define safe startup, shutdown and lifecycle behaviour.
- Establish the local connection and trust boundaries between RoofDeck and Roof.Engine.
- Define initial acceptance tests and regression requirements.

Initial Target Environment: Generic Linux x86-64.

This provides an initial path away from Raspberry Pi-specific operation while retaining the established Linux foundation.

Linux x86-64 is a development target at this stage, not a statement of existing compatibility. Windows and macOS Engine runtime support will be evaluated separately.

Expected Outcome: A documented architectural baseline and approved implementation scope for the first hardware-independent runtime.

Hardware-Independent Runtime Proof of Concept (December)

Target Milestone: Generic-Host Runtime Proof of Concept
Target Period: 18-23 December 2026

The next objective is to demonstrate the existing Roof foundation operating on a general-purpose Linux computer without requiring a dedicated Raspberry Pi or unconditional modification of the host operating system.

Planned Capabilities

- Execute Roof.Engine on a generic Linux x86-64 host.
- Isolate device provisioning and Raspberry Pi-specific management.
- Make Wi-Fi access-point and appliance management functionality optional.
- Support configuration and startup without requiring dedicated-appliance behaviour.
- Preserve Roof identity and configuration across restarts.
- Connect RoofDeck to a locally running Engine.
- Validate local client registration and authentication.
- Exercise existing communication, command exchange and storage functionality.
- Assess existing RApp compatibility.
- Document any functionality that continues to require dedicated-host integration.

Expected Outcome: A functional technical proof of concept demonstrating hardware-independent Roof operation, including verified local functionality and documented limitations.

This milestone is not a general-user release. It establishes a technical foundation for subsequent development.

Q1 2027: Functional Hardware-Independent Roof

Period: January-March 2027

Our objective for Q1 is to turn the generic-host proof of concept into a reliable, minimally privileged software runtime while preserving the capabilities Roof already provides.

The focus will be on adapting and validating the existing implementation rather than introducing unrelated new features.

By the end of Q1, we aim to demonstrate the following workflow on a supported generic host:

Install and configure Roof → Establish a persistent Roof identity → Connect RoofDeck → Create and access data → Communicate with another Roof node → Exercise supported application and sharing workflows

Independent Runtime Alpha: Local Functionality (January)

Target Milestone: Independent Runtime Alpha (Local)
Target Date: 29 January 2027, subject to acceptance testing

This milestone focuses on safe installation, identity, storage and local operation on a supported generic Linux environment.

Planned Capabilities

- Establish repeatable installation, upgrade and removal procedures.
- Support appropriate Engine configuration for a general-purpose host.
- Provide foreground startup and optional background service registration.
- Avoid unnecessary system-wide modification.
- Preserve Roof identity and cryptographic keys across sessions.
- Validate client registration, authentication and credential protection.
- Support appropriate storage isolation without mandatory per-client Linux accounts.
- Validate shutdown, restart and interruption recovery.
- Connect RoofDeck to the locally operating Engine.
- Document supported capabilities and remaining limitations.

Success Criteria: A supported generic Linux host can operate the existing Roof foundation locally without requiring dedicated Raspberry Pi hardware or unsafe appliance-style administration.

The alpha is intended for controlled technical validation rather than unrestricted everyday use.

Independent Runtime Alpha: Remote Connectivity (February)

Target Milestone: Independent Runtime Alpha (Remote)
Target Date: 26 February 2027, subject to acceptance testing

Following local runtime validation, development will focus on communication between independent Roof nodes.

Planned Capabilities

- Operate two independent Roof nodes on non-Raspberry Pi hosts.
- Validate identity and authentication between independent nodes.
- Test existing message and command exchange functionality.
- Establish remote connectivity through Roof.PathFinder.
- Validate communication across physically separate networks.
- Test session establishment, termination and replacement.
- Assess reconnection following temporary network interruptions.
- Improve diagnostics for networking and lifecycle failures.
- Document remote deployment and connectivity limitations.

Success Criteria: Two hardware-independent Roof nodes can establish authenticated communication through supported networking infrastructure, with verified session and recovery behaviour.

This remains a technical alpha, not a consumer beta.

Existing Functionality Integration (March)

Target Milestone: Functional Integration Review

March will focus on verifying that the transition to independent operation has not compromised the functionality already established in Roof.

Planned Work

- Validate existing RApp registration, startup, execution and shutdown.
- Test application data creation, retrieval and persistence.
- Identify remaining application dependencies on dedicated-host filesystem structures or user accounts.
- Validate existing user-to-user data-sharing operations.
- Verify supported access permissions and authorisation.
- Test access revocation where already implemented.
- Improve general-purpose-host configuration, onboarding and error reporting.
- Conduct controlled external technical validation.
- Document functional gaps, security considerations and unsupported workflows.

Success Criteria: The hardware-independent runtime preserves the supported core identity, communication, application, storage and sharing functionality, with any unsupported behaviours explicitly documented.

The Q1 review will determine which capabilities and deployment workflows are ready to progress towards user-facing software beta testing.

Q2 2027: Software-First User Deployment

Period: April-June 2027

Objective: Make Roof usable on supported everyday computing devices, rather than solely executable on development and test hosts.

Q2 marks the transition from hardware-independent technical validation towards a practical software-first user experience.

Software-First Installation and Host Safety Alpha (April)

Our immediate priority is establishing a repeatable and appropriately safe deployment workflow for supported user devices.

Planned Work

- Improve installation, initial configuration and removal procedures.
- Validate safe startup, shutdown and update behaviour.
- Strengthen host-level permission and data isolation boundaries.
- Remove unnecessary requirements for whole-host administration.
- Improve Engine and RoofDeck integration during setup.
- Document supported environments and installation restrictions.
- Test recovery, failure handling and host-system safety.

Expected Outcome: A supported installation and lifecycle workflow suitable for controlled validation on general-purpose Linux devices.

Roof Software Beta 1: Core User Workflows (May)

Target Milestone: Roof Software Beta 1
Release Status: Conditional on readiness

Our first planned user-facing software beta will evaluate whether users can meaningfully operate Roof without purchasing dedicated hardware.

Planned Validation

- Install and configure Roof on a supported user device.
- Establish and persist a Roof identity.
- Connect and operate RoofDeck.
- Create, manage and access personal data.
- Exercise supported RApp functionality.
- Establish communication between independent Roof nodes.
- Validate currently supported data-sharing operations.
- Assess application reliability, configuration and recovery behaviour.

Success Criteria: Participating users can perform supported core Personal Internet workflows through a documented software-first deployment process, without requiring dedicated Roof hardware.

Software-First Beta Validation Review (June)

Following the initial software beta, we will focus on reliability, security and the repeatability of supported deployments.

Focus Areas

- Review installation and configuration results.
- Evaluate host-system safety and privilege boundaries.
- Assess data integrity, authentication and access-control behaviour.
- Identify recurring reliability and connectivity defects.
- Improve software lifecycle and recovery behaviour.
- Consolidate participant feedback and unresolved limitations.
- Determine readiness for broader software beta participation.

Expected Outcome: An evidence-based assessment of Roof's software-first deployment maturity and priorities for the next development phase.

Q3 2027: Platform Portability and Broader Software Beta

Period: July-September 2027

Objective: Expand supported environments only after the existing hardware-independent runtime and its core workflows have been demonstrated to be reliable.

During Q3, development will focus on broader platform compatibility, deployment improvements and external validation.

Platform Portability (July)

Planned Work

- Evaluate additional operating-system and processor-architecture targets.
- Identify remaining Linux-specific dependencies.
- Adapt runtime components for the next approved platform target.
- Preserve shared networking, identity and security contracts.
- Test compatibility between supported Roof deployments.
- Publish a verified support matrix and explicit exclusions.

Windows and macOS Engine runtime support remain potential future targets, not confirmed delivery commitments.

RoofDeck platform support and Roof.Engine runtime compatibility will be assessed separately.

Expanded Software Beta (August)

Indicative Participant Target: 20–50 participants, subject to readiness and support capacity.

Focus Areas

- Validate Roof on an expanded set of supported computing environments.

- Assess installation and onboarding experiences.
- Test communication, application and data workflows.
- Identify compatibility issues across supported deployments.
- Gather reliability and usability feedback.
- Improve documentation and support procedures.

Expansion will depend on installation safety, functional readiness, security and the capacity to support participants.

Public Software Beta Candidate (September)

Target Milestone: Roof Public Software Beta Candidate

The objective is to evaluate whether Roof is sufficiently stable, secure and usable to support wider public beta participation.

Assessment Criteria

- Repeatable installation and configuration.
- Verified support for declared operating systems and architectures.
- Reliability of core user workflows.
- Safe host-system behaviour.
- Identity, credential and data protection.
- Communication and recovery reliability.
- Documented limitations and known defects.
- Availability of appropriate user and developer guidance.

A public beta candidate will be declared only if the relevant acceptance criteria are met.

Q4 2027: Platform Consolidation

Period: October-December 2027

Objective: Consolidate Roof as a sustainable hardware-independent platform while supporting dedicated hardware as an explicitly optional deployment method.

Platform Consolidation (October)

Focus Areas

- Improve runtime stability and performance.
- Consolidate supported host deployment workflows.
- Review software update and lifecycle mechanisms.
- Improve security and diagnostics.
- Maintain interoperability between supported environments.
- Refine developer interfaces and application integration.

Optional Hardware Deployment Validation (November)

Target Milestone: Optional Hardware Validation

With the hardware-independent runtime established, we intend to revisit Roof's original plug-and-play hardware model.

Rather than maintaining a separate required platform, dedicated hardware should operate as an optional deployment profile using the same underlying Roof codebase.

Planned Work

- Validate the existing Raspberry Pi deployment pathway against the portable core.
- Separate optional appliance provisioning from standard Engine operation.
- Test explicitly enabled device-management capabilities.
- Assess compatibility and functionality across software-based and hardware-based deployments.
- Document hardware-specific capabilities and restrictions.

Expected Outcome: A validated approach for optional dedicated-hardware operation that does not make hardware ownership a condition for using Roof.

This milestone does not commit to the production or sale of new hardware.

Annual Platform Review (December)

Target Milestone: Roof 2027 Software-First and Optional-Hardware Platform Review

At the end of 2027, we will review:

- Validate the existing Raspberry Pi deployment pathway against the portable core.
- Separate optional appliance provisioning from standard Engine operation.
- Test explicitly enabled device-management capabilities.
- Assess compatibility and functionality across software-based and hardware-based deployments.
- Document hardware-specific capabilities and restrictions.

The findings will inform Roof's development priorities for 2028.

Beyond the Current Roadmap

Our long-term vision extends beyond the initial hardware-independent runtime. Future research and development may include below.

Decentralised Availability (Buddy System)

The Buddy System is a proposed approach for improving availability when a user's primary Roof environment is offline.

Potential research areas include cooperative node availability, offline communication, recovery and resilient access to personal infrastructure.

Active implementation is deferred while we prioritise hardware independence, software-first deployment and validation of existing functionality.

Future Buddy System development will depend on the maturity of the core platform, technical feasibility and available engineering capacity.

Broader Platform Support
We intend to investigate operating Roof across additional operating systems and computing environments.

However, support will be declared only when the relevant runtime, installation, security and functional requirements have been tested.

Application Ecosystem Development
As the software-first platform matures, we intend to continue improving application integration, developer interfaces and supported Personal Internet workflows.

These initiatives remain longer-term development directions rather than committed deliverables for the current roadmap.

Roadmap Updates

This roadmap describes planned development rather than guaranteed release dates or functionality.

A milestone will be considered achieved only when its stated acceptance criteria have been met.

In particular:

- Source publication does not establish deployment readiness.
- Hardware independence does not automatically establish cross-platform compatibility.
- Successful technical execution does not automatically establish end-user readiness.
- Experimental alpha releases do not imply production suitability.
- Supported features and operating systems will be identified through verified testing rather than planned intentions.

We will update this roadmap as development progresses, including changes to milestones, supported environments and known limitations.

For current implementation details, consult the relevant repository documentation, release notes and issue tracker.